
Validation and Verification document for Cow-puter Vision-Animal Detection on Dairy Farms

Version 1

Team 21

COMPSCI 4ZP6

Dr.Mehdi Moradi

April 7, 2026

Group Members	Student Number	Contact
Xianpi Duan	400381987	duanx14@mcmaster.ca
Xiaozhao Duan	400381647	duanx15@mcmaster.ca
Zicheng Li	400359321	li1276@mcmaster.ca
Bozhi Zhang	400327039	zhanb50@mcmaster.ca
Zhanxu Ye	400390561	yez69@mcmaster.ca

Validation and Verification document for Cow-puter Vision-Animal Detection on Dairy Farms.....	0
V&V plan.....	1
Component Test Plan for Frontend.....	1
Unit Tests.....	1
API (__tests__/api/apiSetup.test.ts).....	1
Services (__tests__/services/).....	1
Hooks (__tests__/hooks/).....	1
Components (__tests__/components/).....	1
Full Unit Tests Coverage Report.....	3
Component Test Plan for Backend.....	5
API Endpoints Unit Test.....	5
/api/setup (SetupView) — 10 tests, all PASSED.....	5
/api/auth (AuthView) — 5 tests, all PASSED.....	6
/api/password-reset — 9 tests, all PASSED.....	7
/api/tracks (TracksView) — 8 tests, all PASSED.....	8
/api/alerts (AlertRuleListView + AlertRuleDetailView) — 14 tests, all PASSED.....	8
/api/settings (SettingsView) — 9 tests, all PASSED.....	9
/api/playback (PlaybackView) — 8 tests, all PASSED.....	10
/api/reports (ReportListView) — 7 tests, all PASSED.....	11
/hls/* and /rec/* (HLSView, RecView, MediaTokenView) — 13 tests, all PASSED.....	11
The Intelligence Daemon.....	12
Unit Tests (Logic & Behavior).....	12
Inference Engine (Mocked Stream) — 13 tests, all PASSED.....	13
VideoStream:.....	13
BehaviorClassifier:.....	14
DatabaseHandler:.....	14
Event Monitor — 16 tests, all PASSED.....	14
StateTracker:.....	14
AlertRuleEngine (cooperates with StateTracker):.....	15
NotificationService:.....	15
Notification Dispatch — 9 tests, all PASSED.....	16
SendEmailAlert:.....	16
SendWebhookAlert:.....	16
DispatchExternal:.....	16
PlaybackManager — 11 tests, all PASSED.....	17
IndexParser:.....	17
MetadataSyncer:.....	17
RetentionService:.....	17
ReportManager — 10 tests, all PASSED.....	18

ReportGenerator:	18
ReportStorage:	18
The Media Server	19
Integration Tests (End-to-End Flow)	19
Component Test Plan for AI model	19
Data Partitioning	19
Component 1: Detection	19
Performance Tests & Metrics	19
Component 2: Behaviour Classification	20
Performance Tests & Metrics	20
Component 3: Tracking	26
Performance Tests & Metrics	26
Component 4: Runtime	28
Performance Tests & Metrics	28

V&V plan

Component Test Plan for Frontend

Unit Tests

Jest (<https://jestjs.io/>) is the main framework that we leveraged to test against our Next.js frontend. The following apps' layers are tested by unit tests:

- Pages
- Components
- Hooks
- Services
- API

API (__tests__/api/apiSetup.test.ts)

Axios interceptor logic is being tested here, including auth header injection, 401 token refresh, and redirect on failure.

Services (__tests__/services/)

6 service modules are being tested here: alert, auth, playback, report, settings, track. The services layer is tested with mocked clients.

Hooks (__tests__/hooks/)

- hooks.modules.test.ts — smoke test that all 8 hooks export as functions.
- query-hooks.behavior.test.ts — wiring between hooks and services.

We mocked `@tanstack/react-query` so `useQuery/useMutation` return their options object rather than real React state.

Components (__tests__/components/)

File	Subjects	Approach
basic-components	CPButton, CPInput, CPTextarea, CPPageHeader, CPModal, RequireAuth, QueryProvider	Render + fireEvent, check classes, overflow, localStorage
CPBrand	CPBrand, CPLogo	Render, inspect className for variant logic
CPVideoWithBBox	Video+canvas DOM structure, hook prop passing	container.querySelector, expect.objectContaining
ConditionQueryBuilder	Pure functions queryToJsonLogic / jsonLogicToQuery + component render	Unit tests on exported functions; react-querybuilder mocked to a <div>
visualizations	CPCowStateTimeline, CPDoughnutPlot, CPStackAreaPlot	Render with empty/loading/data states; echarts mocked to <div data-testid="echarts">
AlertModal	Create/edit modes, tab switching, form submit, delete, email toggle, event expansion	setupMocks() helper with override pattern; jest.spyOn(window, 'confirm')
ReportModal	Loading/empty/data states, duration formatting, close	Simple render checks against formatted text

Pages (__tests__/pages/)

File	Pages	Key Scenarios
root-pages	Home, DashboardRoot, AuthBackground	redirect() called, brand renders
auth-pages	Login, Register, ForgotPassword	Form submit, success callback intercept, token storage, router.push

File	Pages	Key Scenarios
reset-pages	ResetPassword, ResetEmail	URLSearchParams mock to control token presence/absence
dashboard-alerts	AlertsPage	Loading/empty/error/data, open modal button
dashboard-live-camera	LiveCameraPage	Behavior toggles (CSS line-through), slider, cow select, advanced config
dashboard-overview	OverviewPage	Stat cards, alert events summary, multi-hook coordination
dashboard-playback	PlaybackPage	Loading/error/empty/data, segment selection, date input, speed buttons, Play/Pause/Restart/Fullscreen /±10s via HTMLMediaElement property mocking
dashboard-reports	ReportsPage	Table render, row-click opens modal
dashboard-settings	SettingsPage	Tab switching, email reset, success/error messages

Full Unit Tests Coverage Report

Tests: 212 passed, 212 total

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Lines
All files	73.74	73.48	76.16	75.07	
api	95.08	76.19	50.00	94.91	
└─ apiSetup.ts	93.75	76.19	60.00	93.33	16,21
└─ client.ts	96.55	100.00	0.00	96.55	563

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Lines
app	100.00	100.00	100.00	100.00	
└─ page.tsx	100.00	100.00	100.00	100.00	
app/auth	100.00	100.00	100.00	100.00	
└─ auth-background.tsx	100.00	100.00	100.00	100.00	
app/auth/forgot-password	100.00	90.00	100.00	100.00	
└─ page.tsx	100.00	90.00	100.00	100.00	45
app/auth/login	100.00	85.71	100.00	100.00	
└─ page.tsx	100.00	85.71	100.00	100.00	60-69
app/auth/register	100.00	80.00	100.00	100.00	
└─ page.tsx	100.00	80.00	100.00	100.00	46-49
app/dashboard	100.00	100.00	100.00	100.00	
└─ page.tsx	100.00	100.00	100.00	100.00	
app/dashboard/alerts	87.50	89.18	60.00	90.32	
└─ page.tsx	87.50	89.18	60.00	90.32	33,37,104
app/dashboard/live-camera	100.00	97.91	100.00	100.00	
└─ page.tsx	100.00	97.91	100.00	100.00	70
app/dashboard/overview	89.65	91.54	92.00	92.95	
└─ page.tsx	89.65	91.54	92.00	92.95	34-37,174
app/dashboard/playback	92.70	83.52	78.57	98.71	
└─ page.tsx	92.70	83.52	78.57	98.71	209
app/dashboard/reports	95.65	100.00	87.50	95.65	
└─ page.tsx	95.65	100.00	87.50	95.65	25
app/dashboard/settings	80.00	71.73	66.66	81.39	
└─ page.tsx	80.00	71.73	66.66	81.39	46-55,61
app/reset-email	91.30	88.88	80.00	95.23	
└─ page.tsx	91.30	88.88	80.00	95.23	34
app/reset-password	92.00	86.36	83.33	95.65	
└─ page.tsx	92.00	86.36	83.33	95.65	41
components	92.64	93.38	75.00	92.17	
├─ CPBrand.tsx	100.00	100.00	100.00	100.00	
├─ CPButton.tsx	100.00	100.00	100.00	100.00	
└─ CPIInput.tsx	100.00	90.47	100.00	100.00	29-40

Component Test Plan for Backend

Pytest will be used as the primary test framework, along with pytest-django for Django-specific testing utilities. Mocking libraries like unittest.mock will be employed to simulate RTSP streams and FFmpeg processes.

API Endpoints Unit Test

All endpoints: /api/setup, /api/auth, /api/tracks, /api/alerts, /api/settings, /api/playback, /api/reports, /hls/* will be tested using the following unit test procedure:

- Authorization Test:
 - Valid: Request with a valid JWT returns 200 OK (or 201 Created).
 - Invalid: Request with an expired/tampered JWT returns 401 Unauthorized.
 - Missing: Request without headers returns 401 Unauthorized.
- Input Validation Test:
 - Malformed JSON: Send broken JSON; expect 400 Bad Request.
 - Missing Fields: Send payload missing required keys; expect 400 Bad Request.
 - Type Mismatch: Send a string where an integer is expected; expect 400 Bad Request.
- Response Schema Test:
 - Verify the response JSON structure matches the contract exactly.
- Database State Test:
 - For POST/DELETE requests, verify the database row was actually created or deleted using a direct DB query in the test teardown.
- Behavioral Test:
 - Valid: Send valid input data; expect correct processing and response.
 - Invalid: Send invalid input data; expect appropriate error handling and response.

/api/setup (SetupView) — 10 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_get_uninitialized	Behavioral	Returns 200 with initialized=false when AppConfig table is empty	PASSED
2	test_get_initialized	Behavioral	Returns 200 with initialized=true when AppConfig row exists	PASSED
3	test_post_success	Response Schema	First-time setup returns 201 with token and refresh keys	PASSED
4	test_post_creates_app_config	Database State	POST creates an AppConfig DB row with is_initialized=True and correct rtsp_url	PASSED

#	Test Name	Category	Description	Result
5	test_post_creates_user	Database State	POST creates a Django User with the provided username	PASSED
6	test_post_already_initialized	Behavioral	Returns 403 when app is already initialized	PASSED
7	test_post_missing_rtsp_url	Input Validation (Missing Fields)	Returns 400 when rtsp_url is missing	PASSED
8	test_post_missing_password	Input Validation (Missing Fields)	Returns 400 when password is missing	PASSED
9	test_post_missing_email	Input Validation (Missing Fields)	Returns 400 when email is missing	PASSED
10	test_post_invalid_email_format	Input Validation (Type Mismatch)	Returns 400 for a malformed email address	PASSED
11	test_post_no_auth_required	Authorization (Valid)	Setup endpoint is public — no Authorization header needed; response is not 401	PASSED

/api/auth (AuthView) — 5 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_login_success	Authorization (Valid) / Response Schema	Valid credentials return 200 with token and refresh keys	PASSED
2	test_login_invalid_password	Behavioral (Invalid)	Wrong password returns 401	PASSED
3	test_login_nonexistent_user	Behavioral (Invalid)	Non-existent username returns 401	PASSED
4	test_login_missing_username	Input Validation (Missing Fields)	Missing username field returns 400	PASSED
5	test_login_missing_password	Input Validation (Missing Fields)	Missing password field returns 400	PASSED

/api/password-reset — 9 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_request_reset_returns_200_for_known_email	Behavioral (Valid)	Returns 200 and sends an email when the address is registered	PASSED
2	test_request_reset_returns_200_for_unknown_email	Behavioral (Valid)	Returns 200 for unknown email (prevents enumeration); no email sent	PASSED
3	test_request_reset_email_contains_token_link	Response Schema	Sent email body contains a reset link with token= parameter	PASSED
4	test_request_reset_missing_email_returns_400	Input Validation (Missing Fields)	Missing email field returns 400	PASSED
5	test_confirm_reset_with_valid_token	Behavioral (Valid)	Valid token allows setting a new password; DB password updated	PASSED
6	test_confirm_reset_with_expired_token	Authorization (Invalid)	Expired/bad token returns 400	PASSED
7	test_confirm_reset_with_tampered_token	Authorization (Invalid)	Tampered token returns 400	PASSED
8	test_confirm_reset_with_wrong_salt	Authorization (Invalid)	Token signed with wrong salt is rejected (400)	PASSED
9	test_confirm_reset_nonexistent_user	Behavioral (Invalid)	Token referencing a deleted user returns 400	PASSED
10	test_confirm_reset_short_password	Input Validation (Type Mismatch)	New password below minimum length returns 400	PASSED
11	test_confirm_reset_missing_token	Input Validation (Missing Fields)	Missing token field returns 400	PASSED
12	test_old_password_no_longer_works_after_reset	Database State	After reset, old password returns 401; new password returns 200	PASSED

/api/tracks (TracksView) — 8 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_post_no_auth	Authorization (Missing)	Missing Authorization header returns 401	PASSED
2	test_post_tampered_jwt	Authorization (Invalid)	Tampered/invalid JWT returns 401	PASSED
3	test_post_valid_query_returns_list	Behavioral (Valid)	Valid JsonLogic rule returns matching TrackingData records	PASSED
4	test_post_empty_result	Behavioral (Valid)	Valid rule with no matching records returns empty list	PASSED
5	test_post_response_schema	Response Schema	Items contain id, cow_id, behavior, timestamp, bbox	PASSED
6	test_post_time_range_filter	Behavioral (Valid)	Time-range JsonLogic rule returns only records within the range	PASSED
7	test_post_non_dict_body	Input Validation (Malformed JSON)	Non-object body (list) returns 400	PASSED
8	test_post_unsupported_operator	Input Validation (Type Mismatch)	Unknown JsonLogic operator returns 400	PASSED
9	test_post_unknown_variable	Input Validation (Type Mismatch)	Reference to unknown variable returns 400	PASSED

/api/alerts (AlertRuleListView + AlertRuleDetailView) — 14 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_list_alerts_no_auth	Authorization (Missing)	Missing Authorization header on GET returns 401	PASSED
2	test_create_alert_no_auth	Authorization (Missing)	Missing Authorization header on POST returns 401	PASSED
3	test_list_alerts_empty	Behavioral (Valid)	Returns 200 with empty list when no rules exist	PASSED
4	test_list_alerts_returns_rules	Behavioral (Valid)	Returns 200 with list containing all created rules	PASSED

#	Test Name	Category	Description	Result
5	test_create_alert_success	Database State	Valid payload returns 201 and creates a DB row	PASSED
6	test_create_alert_response_schema	Response Schema	Response includes id, name, description, conditions, actions, is_active, last_modified_at	PASSED
7	test_create_alert_missing_name	Input Validation (Missing Fields)	Missing name field returns 400	PASSED
8	test_create_alert_missing_conditions	Input Validation (Missing Fields)	Missing conditions field returns 400	PASSED
9	test_get_alert_detail	Behavioral (Valid)	Returns 200 with the requested alert rule including events	PASSED
10	test_get_alert_not_found	Behavioral (Invalid)	Returns 404 when alert rule does not exist	PASSED
11	test_update_alert_success	Database State	PUT with valid data returns 200 and persists the change	PASSED
12	test_update_alert_not_found	Behavioral (Invalid)	Returns 404 when updating a non-existent rule	PASSED
13	test_delete_alert_success	Database State	DELETE returns 204 and removes the DB row	PASSED
14	test_delete_alert_not_found	Behavioral (Invalid)	Returns 404 when deleting a non-existent rule	PASSED

/api/settings (SettingsView) — 9 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_get_settings_no_auth	Authorization (Missing)	Missing Authorization header on GET returns 401	PASSED
2	test_post_settings_no_auth	Authorization (Missing)	Missing Authorization header on POST returns 401	PASSED
3	test_get_settings_empty	Behavioral (Valid)	Returns 200 with only the user email when no Setting rows exist	PASSED

#	Test Name	Category	Description	Result
4	test_get_settings_returns_dict	Response Schema	Returns 200 with all settings as key-value pairs	PASSED
5	test_update_settings_success	Behavioral (Valid)	Valid dict payload returns 200	PASSED
6	test_update_settings_persisted	Database State	POSTed settings are written to Setting DB rows	PASSED
7	test_update_settings_upserts	Database State	POST updates existing setting without creating duplicates	PASSED
8	test_update_settings_invalid_body	Input Validation (Malformed JSON)	Non-dict body returns 400	PASSED
9	test_reset_settings	Database State	DELETE returns 204 and removes all Setting rows	PASSED

/api/playback (PlaybackView) — 8 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_no_auth	Authorization (Missing)	Missing Authorization header returns 401	PASSED
2	test_missing_start_param	Input Validation (Missing Fields)	Missing start query param returns 400	PASSED
3	test_missing_end_param	Input Validation (Missing Fields)	Missing end query param returns 400	PASSED
4	test_invalid_start_type	Input Validation (Type Mismatch)	Non-integer start param returns 400	PASSED
5	test_invalid_end_type	Input Validation (Type Mismatch)	Non-integer end param returns 400	PASSED
6	test_list_segments_valid_range	Behavioral (Valid)	Returns 200 with segments overlapping the requested range	PASSED
7	test_list_segments_empty_range	Behavioral (Valid)	Returns 200 with empty list when no segments overlap	PASSED

#	Test Name	Category	Description	Result
8	test_response_schema	Response Schema	Items contain filename, start_ts, end_ts	PASSED

/api/reports (ReportListView) — 7 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_no_auth	Authorization (Missing)	Missing Authorization header returns 401	PASSED
2	test_list_reports_empty	Behavioral (Valid)	Returns 200 with empty list when no reports exist	PASSED
3	test_list_reports	Behavioral (Valid)	Returns 200 with all available reports	PASSED
4	test_list_reports_schema	Response Schema	Items contain report_id, report_type, generated_at	PASSED
5	test_get_report_by_id	Behavioral (Valid)	Returns 200 with full report data when querying by report_id	PASSED
6	test_get_report_full_schema	Response Schema	Single-report response includes report_id, report_type, data, generated_at	PASSED
7	test_get_report_not_found	Behavioral (Invalid)	Returns 404 when report_id does not exist	PASSED

/hls/* and /rec/* (HLSView, RecView, MediaTokenView) — 13 tests, all PASSED

#	Test Name	Category	Description	Result
1	test_serve_m3u8	Behavioral (Valid)	GET /hls/<file>.m3u8 returns 200 with correct mpegurl content type	PASSED
2	test_serve_ts_segment	Behavioral (Valid)	GET /hls/<file>.ts returns 200	PASSED
3	test_serve_hls_no_auth_allowed	Authorization (Valid)	HLS endpoint is AllowAny — no Authorization header needed	PASSED
4	test_serve_hls_file_not_found	Behavioral (Invalid)	Returns 404 when requested HLS file does not exist	PASSED

#	Test Name	Category	Description	Result
5	test_serve_hls_path_traversal	Input Validation	Directory traversal path (../etc/passwd) returns 400/404	PASSED
6	test_media_token_no_auth	Authorization (Missing)	Missing Authorization header on /api/media-token returns 401	PASSED
7	test_media_token_generation	Response Schema	Valid request returns 200 with token, url, expires_in	PASSED
8	test_media_token_missing_filename	Input Validation (Missing Fields)	Missing filename query param returns 400	PASSED
9	test_rec_view_no_auth	Authorization (Missing)	GET /rec/<file> without credentials returns 401	PASSED
10	test_rec_view_with_jwt	Authorization (Valid)	GET /rec/<file> with valid JWT returns 200	PASSED
11	test_rec_view_file_not_found	Behavioral (Invalid)	Returns 404 when requested recording file does not exist	PASSED
12	test_rec_view_with_signed_token	Authorization (Valid)	GET /rec/<file>?token=<signed> returns 200 without Authorization header	PASSED
13	test_rec_view_invalid_signed_token	Authorization (Invalid)	GET /rec/<file>?token=<invalid> returns 403	PASSED
14	test_rec_view_range_request	Behavioral (Valid)	HTTP Range request returns 206 Partial Content	PASSED

The Intelligence Daemon

Unit Tests (Logic & Behavior)

- Inference Engine (Mocked Stream):
 - Please refer to the component test plan for the ML model.
- Event Monitor:
 - AlertRuleEngine:
 - Test: Cooperate with StateTracker to simulate cow behavior sequences that should and should not trigger alerts.

- Assertion: Verify alerts are triggered only when conditions are met, including duration checks.
 - NotificationService:
 - Test: Mock alert dispatching (e.g., email sending) when an alert is triggered.
 - Assertion: Verify the dispatch method is called with correct parameters.
- PlaybackManager:
 - IndexParser:
 - Test: Feed a sample index.m3u8 with a specific #EXT-X-PROGRAM-DATE-TIME tag.
 - Assertion: Verify the parser extracts the millisecond-precision Unix timestamp correctly.
- RetentionService:
 - Test: Populate the recordings table with entries older than the retention threshold.
 - Assertion: Verify old entries and corresponding files are deleted after running the service.

Inference Engine (Mocked Stream) — 13 tests, all PASSED

VideoStream:

#	Test Name	Description	Result
1	test_read_returns_frame_and_timestamp	read() returns the stored frame (480×640×3) and timestamp without starting a thread	PASSED
2	test_read_returns_none_when_no_frame	read() returns (None, 0.0) before any frame is captured	PASSED
3	test_read_returns_copy_of_frame	read() returns a copy — mutating the returned frame does not affect the stored frame	PASSED
4	test_is_alive_false_before_start	is_alive() returns False before start() is called	PASSED
5	test_is_alive_false_after_stop	is_alive() returns False after stop() is called	PASSED
6	test_context_manager_calls_stop	__exit__ calls stop()	PASSED
7	test_reconnects_on_stream_failure	_update() reconnects when read() returns False; eventually captures a frame	PASSED

BehaviorClassifier:

#	Test Name	Description	Result
8	test_detection_dataclass_has_required_fields	Detection dataclass has track_id, bbox, behavior, confidence, timestamp	PASSED
9	test_classify_returns_detection_list	classify() returns a list of Detection objects with correct track IDs and bbox	PASSED
10	test_classify_empty_frame_no_detections	classify() returns empty list when YOLO finds no boxes	PASSED
11	test_classify_filters_small_bbox	classify() skips detections whose bounding box area is below min_bbox_area	PASSED
12	test_update_runtime_config_changes_threshold	update_runtime_config() updates confidence threshold, detection classes, and min area	PASSED

DatabaseHandler:

#	Test Name	Description	Result
13	test_write_batch_calls_bulk_create	write_batch() calls TrackingData.objects.bulk_create with 3 objects	PASSED
14	test_write_batch_empty_list_does_not_call_bulk_create	write_batch([]) does not call bulk_create	PASSED
15	test_write_batch_retries_on_database_error	write_batch() retries up to max_retries (3) on DatabaseError	PASSED
16	test_write_batch_timestamps_converted_to_ms	write_batch() converts epoch-seconds timestamps to milliseconds in DB	PASSED

Event Monitor — 16 tests, all PASSED

StateTracker:

#	Test Name	Description	Result
1	test_updates_behavior_on_new_detection	update() stores the cow's current behavior	PASSED
2	test_first_update_sets_start_timestamp	First update() sets behavior_start_ts to the received timestamp	PASSED

#	Test Name	Description	Result
3	test_tracks_cumulative_duration	duration_seconds reflects time between first and last update (~60s)	PASSED
4	test_resets_duration_on_behavior_change	duration_seconds resets to ~0 when behavior transitions	PASSED
5	test_tracks_multiple_cows_independently	update() tracks each cow's state independently	PASSED
6	test_get_states_returns_all_cows	get_states() returns shallow copy of all cow states	PASSED
7	test_get_state_unknown_cow_returns_none	get_state() returns None for an unknown cow_id	PASSED

AlertRuleEngine (cooperates with StateTracker):

#	Test Name	Description	Result
8	test_triggers_alert_when_duration_threshold_met	Alert fires when behavior duration (60s) exceeds configured threshold (30s)	PASSED
9	test_no_alert_when_duration_below_threshold	Alert does not fire when duration (30s) is below threshold (120s)	PASSED
10	test_evaluates_multiple_rules_independently	Each active rule evaluated separately — Feeding Rule matches cow_1, Lying Rule matches cow_2	PASSED
11	test_inactive_rule_not_evaluated	Inactive rules (is_active=False) are not loaded or evaluated	PASSED

NotificationService:

#	Test Name	Description	Result
12	test_dispatch_creates_alert_event_in_db	dispatch() creates an AlertEvent record in the database	PASSED
13	test_dispatch_stores_correct_cow_id_in_details	dispatch() stores cow_id in AlertEvent.details	PASSED
14	test_dispatch_stores_behavior_and_duration	dispatch() stores behavior and duration_seconds in AlertEvent.details	PASSED

#	Test Name	Description	Result
15	test_dispatch_respects_dedup_window	dispatch() does not fire again within the dedup window (returns False, no new event)	PASSED
16	test_dispatch_fires_after_dedup_window_expires	dispatch() fires again after the dedup window has passed (2 events total)	PASSED

Notification Dispatch — 9 tests, all PASSED

SendEmailAlert:

#	Test Name	Description	Result
1	test_sends_email_with_correct_args	send_mail is called with subject containing rule name, body containing cow_id and behavior, correct recipient	PASSED
2	test_logs_warning_on_failure	Failed email does not raise; "Failed to send email" warning is logged	PASSED

SendWebhookAlert:

#	Test Name	Description	Result
3	test_posts_json_payload	urlopen is called with a POST request containing JSON payload with rule and cow_id	PASSED
4	test_logs_warning_on_failure	Failed webhook does not raise; "Failed to send webhook" warning is logged	PASSED

DispatchExternal:

#	Test Name	Description	Result
5	test_dispatches_email_when_configured	Email channel invoked when actions contains email; webhook not called	PASSED
6	test_dispatches_webhook_when_configured	Webhook channel invoked when actions contains webhook; email not called	PASSED
7	test_dispatches_both_channels	Both channels fire when both keys are present	PASSED
8	test_noop_when_actions_empty	No channels invoked when actions is an empty dict	PASSED

#	Test Name	Description	Result
9	test_noop_when_actions_none	No channels invoked when actions is None	PASSED

PlaybackManager — 11 tests, all PASSED

IndexParser:

#	Test Name	Description	Result
1	test_parse_program_date_time_tag	Parses #EXT-X-PROGRAM-DATE-TIME:2026-01-22T13:36:52.000Z and extracts filename + duration	PASSED
2	test_parse_returns_unix_ms_precision	Parsed start_ts_ms is a millisecond Unix epoch integer matching expected value	PASSED
3	test_parse_end_ts_equals_start_plus_duration	end_ts_ms == start_ts_ms + duration * 1000	PASSED
4	test_parse_missing_tag_returns_empty_list	Returns empty list when no segment entries are found	PASSED
5	test_parse_nonexistent_file_returns_empty_list	Returns empty list when file does not exist	PASSED
6	test_parse_multiple_segments	Parses multiple segments from a playlist with 2 entries	PASSED

MetadataSyncer:

#	Test Name	Description	Result
7	test_new_ts_file_creates_video_segment_record	_sync_segment() creates a VideoSegment DB row with correct start_ts	PASSED
8	test_duplicate_file_not_inserted_twice	Calling _sync_segment() twice does not create a second row (count == 1)	PASSED

RetentionService:

#	Test Name	Description	Result
9	test_deletes_segments_older_than_retention_days	_cleanup_by_age() deletes DB rows and files older than 30-day threshold (60-day-old segment)	PASSED
10	test_does_not_delete_recent_segments	_cleanup_by_age() leaves 5-day-old segments untouched	PASSED
11	test_db_record_removed_with_file	_cleanup_by_age() removes both the DB row and the physical file	PASSED
12	test_deletes_oldest_when_disk_exceeds_limit	_cleanup_by_disk() removes oldest-first when usage exceeds disk limit	PASSED

ReportManager — 10 tests, all PASSED

ReportGenerator:

#	Test Name	Description	Result
1	test_aggregates_behavior_counts_correctly	generate_daily_report() counts 2 feeding + 1 standing correctly	PASSED
2	test_computes_total_duration_per_behavior	behavior_summary includes estimated_duration_seconds ≥ 0 for each behavior	PASSED
3	test_empty_data_produces_valid_report_structure	Returns valid structure with total_cows=0, total_records=0 when no data exists	PASSED
4	test_report_includes_all_cows	per_cow breakdown includes all 3 detected cows; total_cows == 3	PASSED
5	test_report_date_field_matches_requested_date	date field matches 2026-01-22	PASSED
6	test_behavior_percentages_sum_to_100	Percentages across all behaviors sum to ~100%	PASSED

ReportStorage:

#	Test Name	Description	Result
7	test_saves_report_creates_db_row	save() creates a Report record in the database	PASSED

#	Test Name	Description	Result
8	test_report_has_correct_metadata_fields	Saved Report has report_id, report_type, data, generated_at	PASSED
9	test_saves_correct_data_payload	Stored report data matches input dict exactly	PASSED
10	test_multiple_reports_stored_independently	Two save() calls create 2 independent DB rows	PASSED

The Media Server

Integration Tests (End-to-End Flow)

- Test: Start the FFmpeg process with a test RTSP stream (could be a local file served via RTSP).
- Assertion: Verify that the HLS files (.m3u8 and .ts) are generated in the expected directory with correct naming conventions.

#	Test Name	Description	Result
1	test_live_feed_generates_hls_files	Spawns FFmpeg with a local .ts file, verifies live.m3u8 is created with #EXTM3U and #EXT-X-TARGETDURATION headers, and at least one .ts segment is generated	PASSED
2	test_hls_segment_naming_convention	Verifies .ts segment files are generated with correct naming and all are non-empty	PASSED

Component Test Plan for AI model

Data Partitioning

To prevent overfitting, data was split 70/10/20. We tuned the model using the validation set and reserved the test set exclusively for the final accuracy assessment.

Component 1: Detection

Performance Tests & Metrics

This component focuses on two primary error types:

- Missed Detection (False Negative): A cow is present in the frame but not detected.
- False Detection (False Positive): A detection box appears where no cow exists.

Metric 1: Intersection over Union (IoU)

IoU measures the overlap between the predicted bounding box and the ground-truth bounding box. B_p Mean predicted bounding box, B_{gt} mean ground-truth bounding box

$$IoU = \frac{Area(B_p \cap B_{gt})}{Area(B_p \cup B_{gt})}$$

Higher IoU indicates better localization accuracy.

IoU will be computed using NumPy.

Metric 2: Precision and Recall

- **Recall** measures the proportion of actual cows that are successfully detected; low recall indicates many missed cows.
- **Precision** measures the proportion of detected boxes that correspond to real cows; low precision indicates many false detections.

Precision and recall will be computed using Ultralytics validation tools or sklearn.metrics (e.g., model.val(), precision_score, recall_score).

Component 1 Result (Images tested: 1907):

True Positives	False Positives	False Negatives
20878	353	441

Precision	Recall	Mean IoU
0.9834	0.9793	0.9153

Component 2: Behaviour Classification

Performance Tests & Metrics

Metric 1: Confusion Matrix

The confusion matrix compares ground-truth behaviors with predicted behaviors.

It helps identify systematic confusion patterns, such as drinking being misclassified as feeding due to proximity to feeding areas.

This analysis reveals whether errors are caused by class imbalance or specific visual scenarios.

The confusion matrix will be computed using Ultralytics or `sklearn.metrics.confusion_matrix` and visualized as a heatmap with Matplotlib.

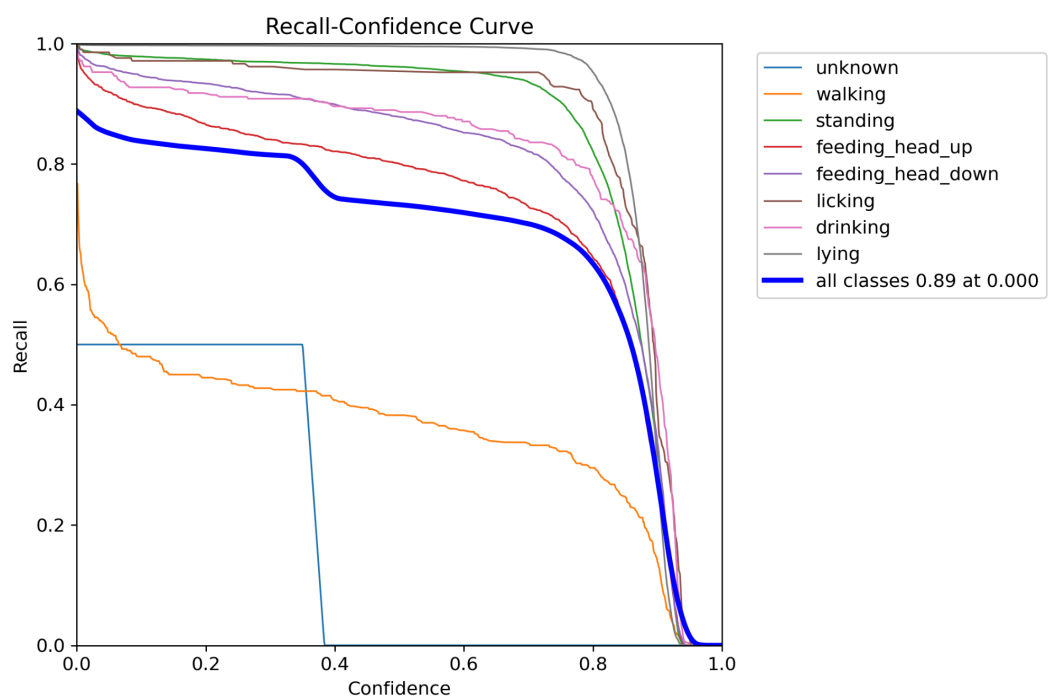
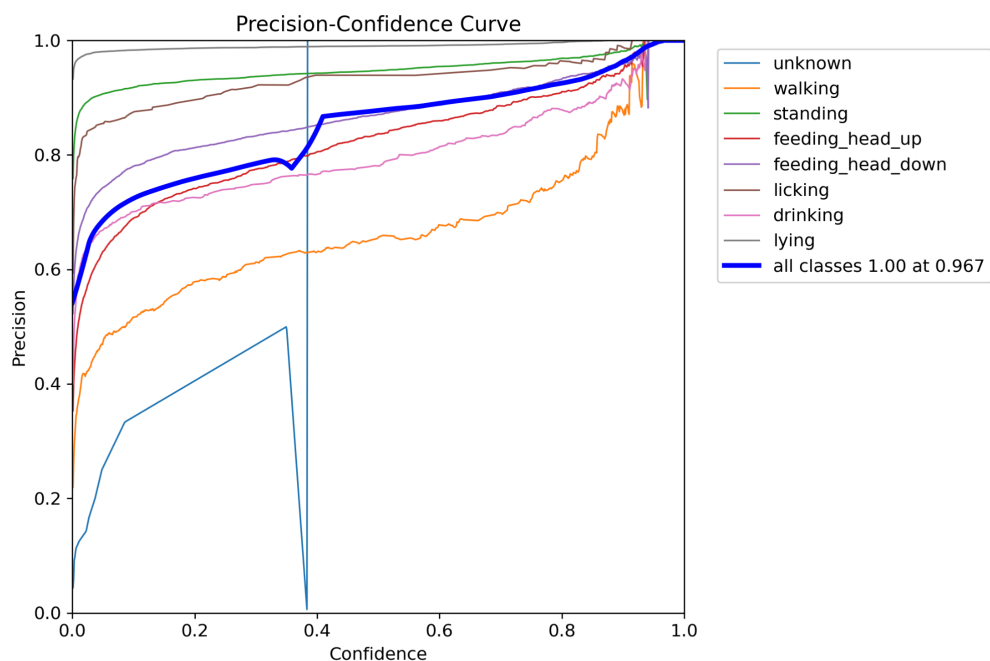


The confusion matrix shows the main confusions are between `feeding_head_up` and `feeding_head_down`, and between `walking` and `standing`, which is expected since these behaviours look visually similar.

Metric 2: Precision and Recall

- Recall measures how many actual occurrences of a behavior are correctly identified; low recall indicates frequent missed detections.
- Precision measures how often predicted behavior labels are correct; low precision indicates frequent false alarms.

The computation of precision and recall follows the same definitions as in the Detection component.



Class	Precision	Recall
unknown	0.4712	0.5
walking	0.6118	0.4255
standing	0.9372	0.9697

feeding_head_up	0.7761	0.8403
feeding_head_down	0.8344	0.9164
licking	0.9224	0.9619
drinking	0.7491	0.9085
lying	0.9878	0.9965

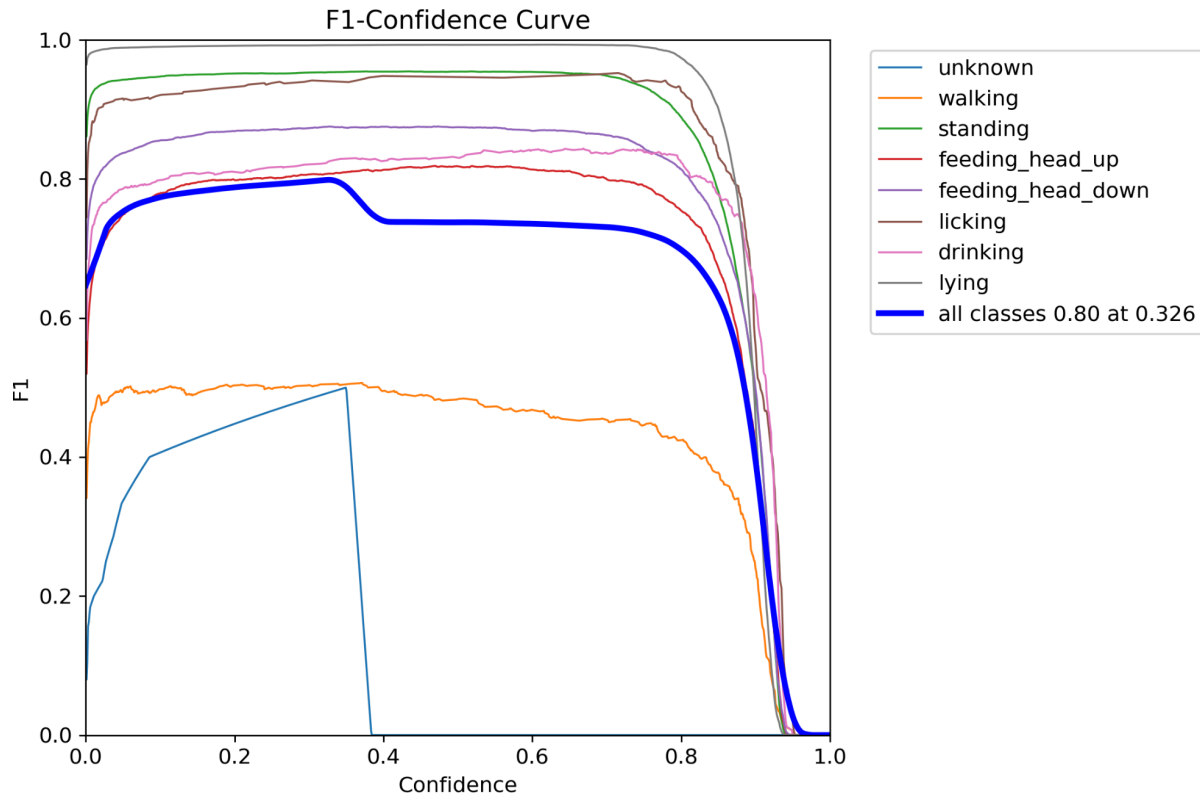
Mean Precision	Mean Recall
0.7862	0.8148

Metric 3: F1 Score

The F1 score balances precision and recall when neither should be favored.

A low F1 score for a behavior indicates unreliable recognition due to either frequent misses or frequent false positives.

The F1 score will be computed using Ultralytics tools or `sklearn.metrics.f1_score`



Class	F1-score
unknown	0.4852
walking	0.5019
standing	0.9532
feeding_head_up	0.8069
feeding_head_down	0.8734
licking	0.9417
drinking	0.8211
lying	0.9922

Mean F1-score: 0.7970

The F1 shows lying, standing, and licking perform the best, while walking and unknown are much lower due to visual similarity with other classes and limited samples. Overall, the model is reliable for the main behaviours.

Metric 4: ROC–AUC (Area Under the Receiver Operating Characteristic Curve)

ROC–AUC evaluates the model’s ability to distinguish a target behavior from all other behaviors across different classification thresholds.

- TPR (True Positive Rate / Recall) = $TP / (TP + FN)$, measuring the model’s ability to correctly identify positive samples.
- FPR (False Positive Rate) = $FP / (FP + TN)$, measuring the rate at which negative samples are incorrectly classified.

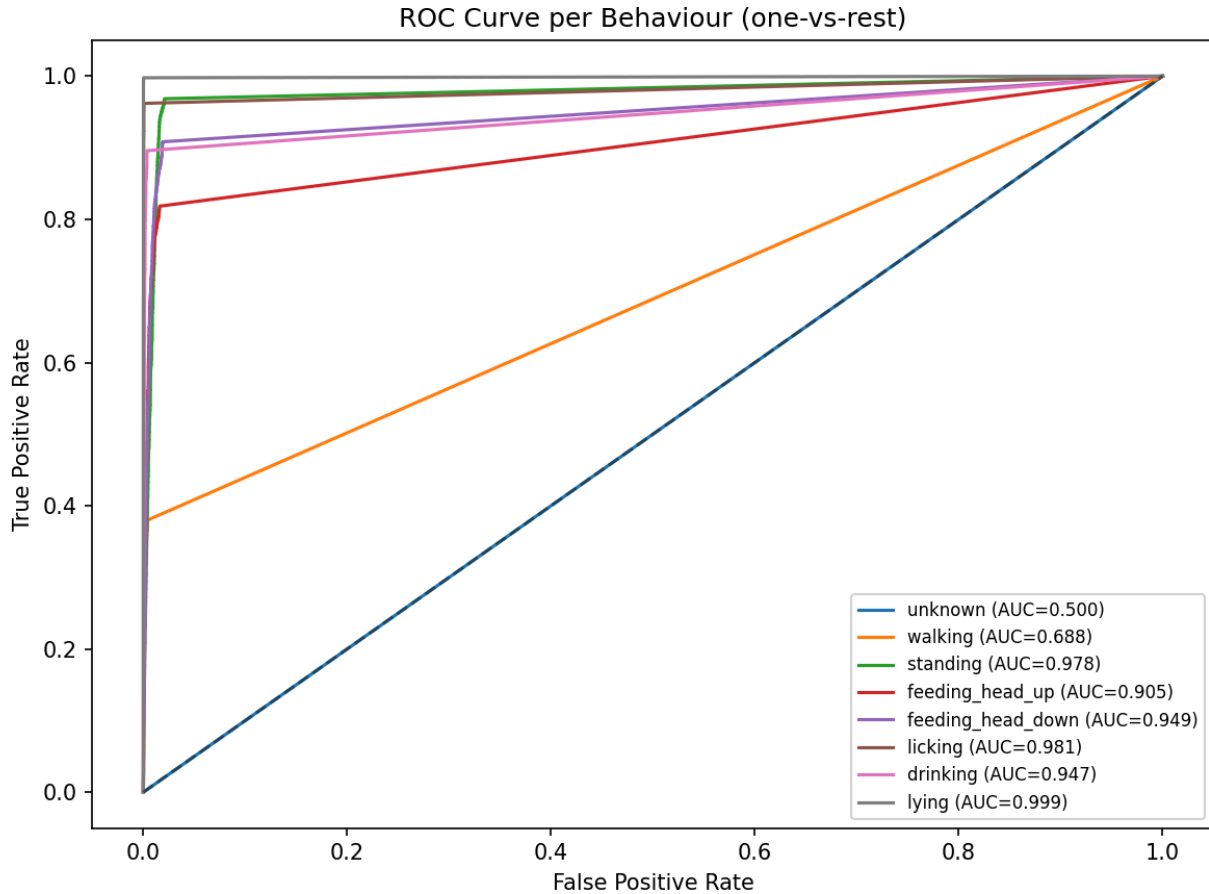
The evaluation procedure is as follows:

1. For each behavior class k , treat k as the positive class and all other behaviors as the negative class.
2. Use `sklearn.metrics.roc_curve` to generate the ROC curve and `sklearn.metrics.auc` to compute the area under the curve.

This produces an AUC score for each behavior class.

Class	ROC-AUC
unknown	0.5
walking	0.6883
standing	0.9777
feeding_head_up	0.9046
feeding_head_down	0.9488

licking	0.9808
drinking	0.947
lying	0.9988



The model achieves high ROC-AUC > 0.90 on most behaviours, indicating strong discriminative ability. But the walking is low with 0.688 due to visual similarity with standing, and unknown is undefined because no positive samples exist in the test set.

Component 3: Tracking

Performance Tests & Metrics

Metric: Ghost Tracks

Ghost tracks are used to detect false positives where non-existent cows are assigned a track ID for a short duration.

For each track_id, we count the number of frames in which it appears.

Tracks that appear for fewer than 5 frames are classified as ghost track candidates.

Metric	Value
Total Frames Seen	1500
Total Tracks	38
Ghost Threshold	< 5 frames
Ghost Tracks	12
Normal Tracks	26
Ghost Ratio	31.58%

Ghost Tracks with Duration < 5 frames	
Track ID	Frames Survived
20	1
22	3
27	1
28	1
31	1
35	3
44	1
54	3
62	2
81	1
84	1

Normal Tracks with Duration $\geq$ 5 frames	
Track ID	Frames Survived
1	1500
2	1498
3	1496
4	1498
5	13
6	6
7	1483

8	1497
9	841
10	852
11	1499
19	283
41	42
48	8
49	101
56	175
64	110
67	162
77	10
78	7
87	168
91	465
92	463
93	443
94	440
104	109

Overall, the tracker works stably. The main cows like ID with 1, 2, 3, 4, 7, 8, 11, etc. are tracked through almost the entire 1500 frames, and the ghost tracks are all very short-lived and do not affect the final result.

Component 4: Runtime

Performance Tests & Metrics

We evaluate runtime performance by measuring the total inference time of the model using Python timing utilities.

During measurement, the start time t_0 , is recorded using `time.perf_counter()`, inference is executed on the video, and the end time t_1 , The total inference time is computed as $T = t_1 - t_0$. Let:

- N be the total number of frames processed during the timed run.
- T be the total inference time.

$$FPS = \frac{N}{T}$$

The runtime performance is measured as frames per second (FPS):

To improve reliability, the experiment is repeated k times and the final average FPS is reported as:

$$\overline{FPS} = \frac{1}{k} \sum_{i=1}^k FPS_i$$

This metric is used to verify whether the model meets the runtime performance requirements.

Parameter and Metric	Value
Hardware (GPU)	NVIDIA H100
Model	YOLOv12
Total Frames	1500
Test Repeats	5 Runs
Average FPS	73.8
Min FPS	73.11
Max FPS	75.79
Average Runtime	20.328 seconds

The model processes 1500 frames in about 20.3 seconds per run, and the average speed is 73.80 FPS on GPU. Across 5 repeated runs, the FPS stays between 73.11 and 75.79, which means the performance is very consistent and there is no big fluctuation.

Since real-time video or CCTV is usually 30 FPS, the model runs more than 2x faster than real-time, so it easily meets the runtime requirement.